

Identifying Financially Constrained Firms: Data

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Outline

Financing Frictions: Empirical Challenges

The object of interest

The regression we would want to estimate:

$$Y_{i,t} = \beta \textit{Frictions}_{i,t} + \alpha_i + \delta_t$$

Where $Y_{i,t}$ is employment, investment, patents, etc.

Assume we can measure $\textit{Frictions}_{i,t}$, what problem remains?

Two broad approaches

1. Find “exogenous shifter” for $Frictions_{i,t}$

- Δ collateral value, Δ working capital, Δ credit supply, access to foreign capital, Δ cash-flow

2. Time-series shocks interacted with ex-ante measures of financing frictions

- Leverage, dividend payment, access to bond market, cash, 10-K, credit lines

3. Mix of both

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!! papers often don't report relevant economic elasticities but just “reduced form IV”

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Brief overview of exogenous shifters: firm level

- Collateral
 - Value of underlying asset (Gan, 2007; Chaney, Sraer, Thesmar, 2012)
 - Typical elasticity: [0.06–0.12]
- Working capital
 - Δ trade credit payment delays (Beaumont, Lenoir, 2025); access to working capital loans (Matray, Mueller, Xu, Kabir, 2025)
 - Typical elasticity: \$1 wc = \$4-\$5 of extra revenues
- “Omnibus” bank credit supply shock + bond credit supply shock (Coppola, 2025)
 - Typical elasticity credit to K or Y ≈ 0.3 (0.26 [Cingano, Manaresi, Sette, 2016]; 0.36 [Amiti, Weinstein, 2018]; 0.23 [Pinardon-Touati, 2025]; 0.31 [Duquero, Matray, Saidi, 2025])
 - Credit to L elasticity typically smaller / inexistant (exception: Chodorow-Reich, 2014; Quincy, 2024)
- Cash-flow or cash-in-hand shocks
 - Multi-segment firms 0.12 [Lamont, 1997], pension shock? 0.6 [Rauh, 2006; Bakke, Whited, 2012], exchange rate 0.3 [Adams, Verdelhan, 2025]

Brief overview of exogenous shifters: “macro” level

- Bank deregulation across the US
 - First wave (Quincy, Xu, 25), second wave?, third wave?
- Collateral / bankruptcy reform: **India** (Vig, 13); **Brazil** (Fonseca, VonDoornik, 22); **US** (Favara et al., 21); **Cross-countries** (Haselmann, Pistor, Vig, 10)
- Foreign capital liberalization (Bau, Matray, 23)
- Bank or branch entry (Fonseca, Matray, 24; Quincy, Xu, 25)

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Identifying financing frictions: ex-ante sorting

Typical equation:

$$Y_{i,t} = \beta \textit{Frictions}_{i,t_0} \times \textit{Shock}_{i,t} + \alpha_i + \delta_t + \dots$$

Identifying financing frictions: ex-ante sorting

Typical equation:

$$Y_{i,t} = \beta Frictions_{i,t_0} \times Shock_{i,t} + \alpha_i + \delta_t + \dots$$

Three challenges:

1. Unless $Shock_{i,t}$ is aggregate, eq. is a triple-diff $\rightarrow$ likely wrongly specified
2. $Frictions_{i,t_0}$ potentially correlated with other stuff
3. Many proxies of financing frictions potentially measures the absence of financing frictions

Who is financially constrained?

1. High cash or low cash?
2. High leverage or low leverage?
3. With more undrawn credit lines or less undrawn credit lines?
4. Large (productive) exporting firms or small domestic firms?
5. Paying dividends or not paying dividends?

As cash increases, firms are less financially constrained

Begenau and Palazzo (2021). Firm selection and corporate cash holdings. *Journal of Financial Economics*

- Cash holding has increased a lot

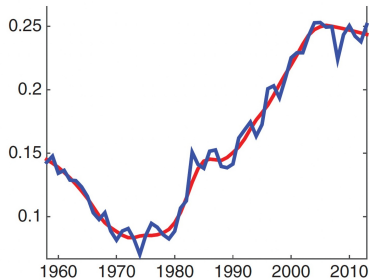


Fig. 1. Average cash-to-assets ratio of US-listed firms. This figure reports the average cash-to-assets ratio of US public companies over the period 1958-2013.

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Begenau and Palazzo (2021). Firm selection and corporate cash holdings. *Journal of Financial Economics*

- R&D firms are **less** financially constrained?

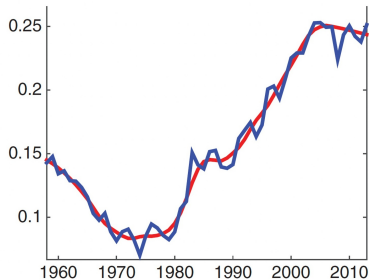


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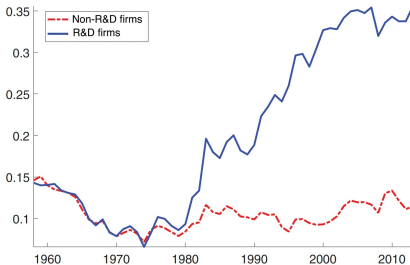


Fig. 3. Average cash-to-assets ratio of US listed firms by sector. This figure reports the average cash-to-assets ratio of R&D-intensive and non-R&D-intensive firms over the period 1958-2013. An R&D-intensive firm belongs to an industry (three-level digit SIC code) whose average R&D investment amounts to at least 2% of assets over the sample period.

As cash increases, firms are less financially constrained

Begenau and Palazzo (2021). Firm selection and corporate cash holdings. *Journal of Financial Economics*

→ Young R&D firms are **less** financially constrained?

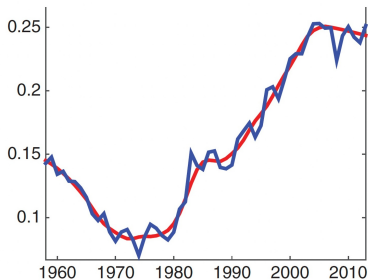
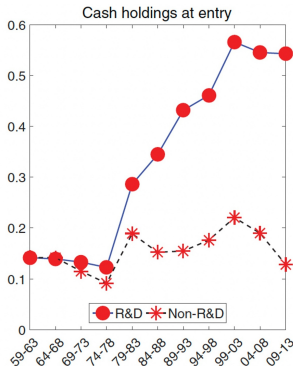


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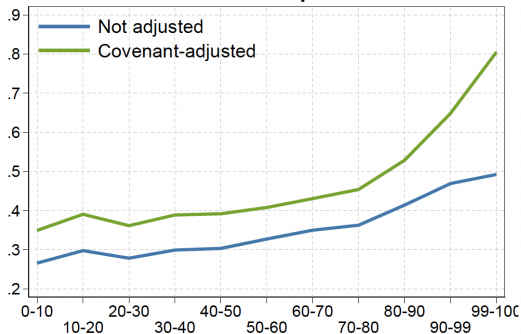


Firms with more slack on their credit lines are less financially constrained

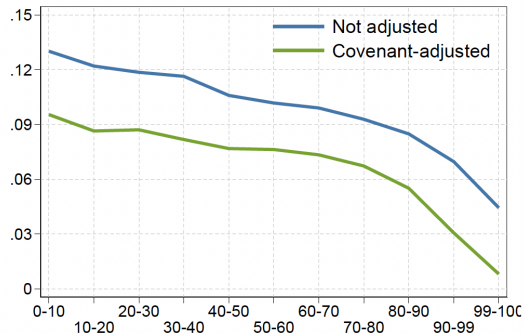
(Amberg, Jacobson, Quadrini, Picco (2023); Aydin, Kim (2026))

→ Large firms are **more** constrained?

C. Max utilization rate in past three months



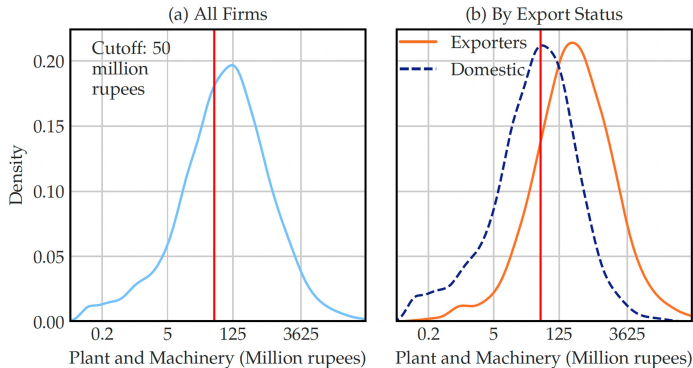
D. Undrawn amount over assets



Exporters are more productive and bigger so should be less constrained

Exporters more productive and bigger $\Rightarrow$ should be less financially constrained (Finlay, 25)

Figure 1: Plant and Machinery Distribution

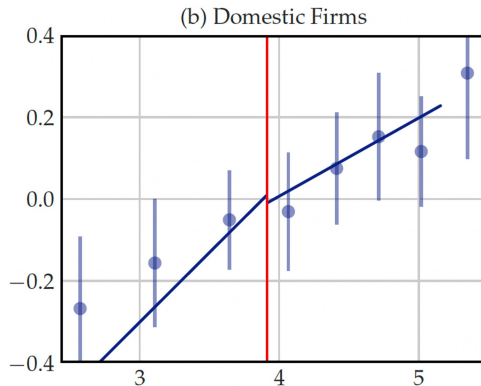
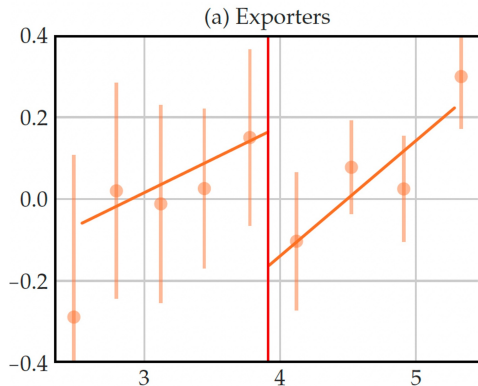


Exporters are more productive and bigger so should be less constrained

Exporters **more productive** and **bigger** $\Rightarrow$ should be **less** financially constrained (Finlay, 25)

Yet they react **more** to a credit shock Xu, 2022; Finlay, 2025; Beaumont, Lenoir, 2025; Matray et al., 2025

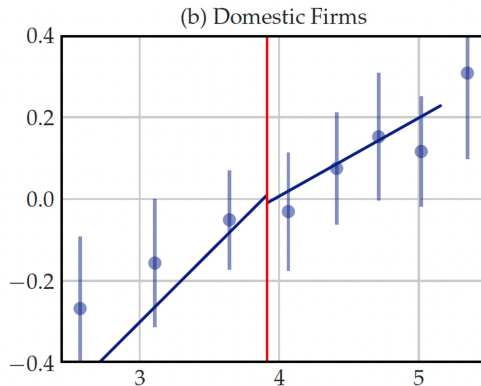
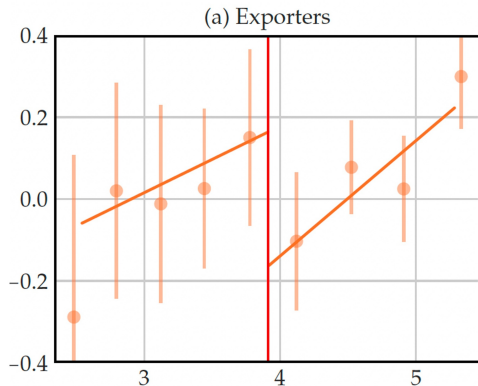
Log(loans)



Exporters are more productive and bigger so should be less constrained

Exporters **more productive** and **bigger** $\Rightarrow$ should be **less** financially constrained (Finlay, 25)
 $\rightarrow$ Financing frictions is the **gap** between **productivity** and **borrowing capacity**

Log(loans)



Where does that leave us?

- Checking [what firms say](#) in their 10k: Hoberg-Maskimovic (2015)

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- Identifying firms with **refinancing needs**: Almeida, Campello, Laranjeira, Weisbenner (2012). Corporate Debt Maturity and the Real Effects of the 2007 Credit Crisis. *Critical Finance Review*

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- Checking **what firms say** in their 10k: Hoberg-Maskimovic (2015)
- Identifying firms with **refinancing needs**: Almeida, Campello, Laranjeira, Weisbenner (2012). Corporate Debt Maturity and the Real Effects of the 2007 Credit Crisis. *Critical Finance Review*
- Sorting firms on their **marginal revenue product of capital** (Bau-Matray, 2023; Matray-Mueller-Xu-Kabir, 2026)

Using MRPK to identify constrained firms

From Lecture 1 (Financing frictions: Theory):

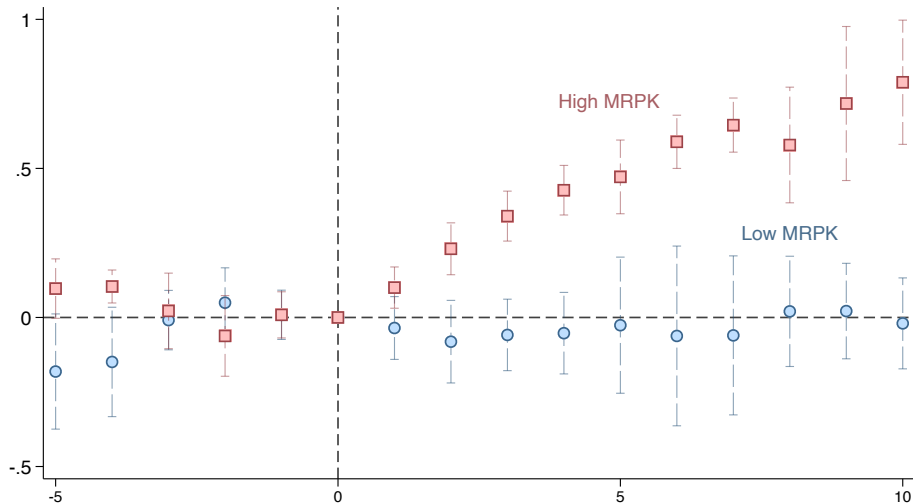
$$MRPK_i = (1 + \tau_i) r$$

τ_i : reduced-form input-cost wedge = degree of financing frictions ($\tau_i = 0$ unconstrained, $\tau_i > 0$ constrained).

- Same user cost r across firms $\Rightarrow$ $MRPK_i$ ranks firms by their wedge τ_i
- High $MRPK \Rightarrow$ high $\tau \Rightarrow$ more constrained (capital too scarce at that firm)
- Dispersion in $MRPK$ = dispersion in financing frictions

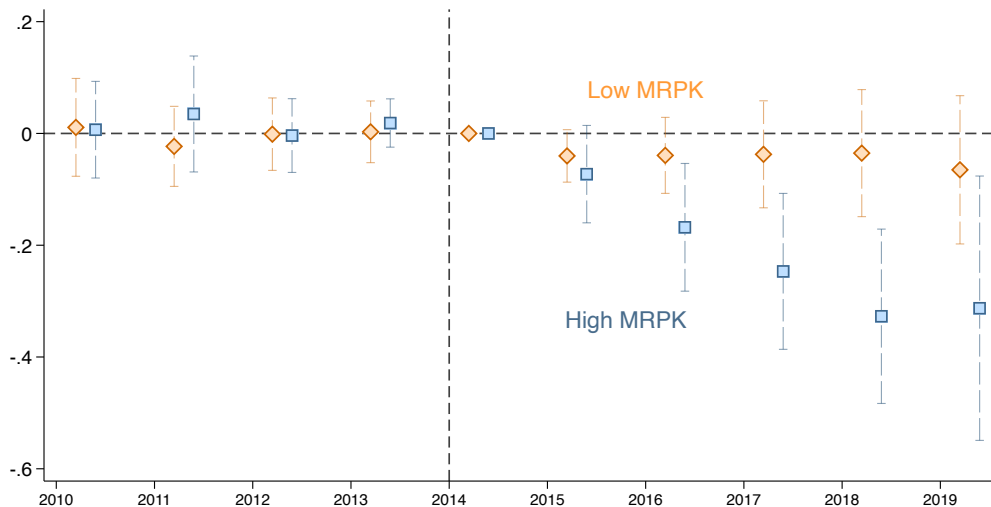
MRPK example: Bau Matray (2023)

Bau Matray (2023): Positive shock



MRPK example: Matray Mueller Xu Kabir (2026)

Matray Mueller Xu Kabir (2026): Negative shock



What to take away

Key takeaway

Financing frictions are the **gap** between a firm's **productivity** and its **borrowing capacity**

- Most single balance-sheet proxies (cash, leverage, size, credit lines) can point the **wrong way**
- Sort on the **gap**: 10-K text, refinancing needs, or the marginal revenue product of capital

Thank you!

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